Structural Theorems for de Sitter–Unruh Modified Inertia: What the First-Moment Closure Forbids

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> **v2 CORRECTION NOTICE — read before §9.** v1's Proposition 7 attached an observational forecast to its two propositions: "~1.2–1.9 on 40–60 Local Group dwarfs, 3 at $N \sim 150$, both routes archival." **That forecast was wrong and is retracted here.** Running the test on the real McConnachie (2012) catalogue shows it is **systematics-limited, not sample-limited**: the per-object scatter is 0.38–0.48 dex rather than the assumed 0.15–0.20, and the dominant error — the stellar mass-to-light ratio — is **coherent** across the sample, so $\sqrt{N}$ does not reduce it. **Propositions 7.1 and 7.2 themselves are unchanged** (a theorem and a calculation); only the forecast built on them fails. §9 is rewritten accordingly. No other section is affected.

Abstract

The modified-inertia action

$$S = \frac{c^4}{16\pi G}\int\sqrt{-g}\,R \;+\; \dots \;-\; \frac{1}{2}\int\sqrt{-g}\,\rho_m\Big[s\,u^\mu K(\Box_u/a_0^2)u_\mu\Big], \qquad K(z) = \frac{\sqrt{1+4z}-1}{2\sqrt{z}},$$

with $\Box_u f = u^a\nabla_a(u^b\nabla_b f)$ and acceleration scale $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$, is evaluated in practice through its **first-moment closure**, $K(\Box_u/a_0^2) \to K(|a|^2/a_0^2) = \mu_{\rm fw}(|a|/a_0)$. We prove seven structural results about that closure. Five are prohibitions and two are predictions.

(1) The first-moment identity $u_\mu\Box_u u^\mu = -|a|^2$ holds **pointwise on every timelike worldline**, following from $u\cdot u = -1$ alone — not merely on the circular and helical cases previously verified. **(2)** The closure's argument is therefore $|a|^2 \geq 0$, while K's branch cut requires $z \leq -\frac{1}{4}$; hence $\mathrm{Im}\,K \equiv 0$ **identically**, for the mean motion and for every perturbation at every order. The MOND amplitude and any dissipative channel are therefore **mutually exclusive**, and the sign postulate s is demoted from a parameter awaiting data to a label on a branch the working theory never evaluates. **(3)** The law $\mu_{\rm fw}(|a|/a_0)\,a = g_{\rm bar}$ is second order, whereas the Euler–Lagrange equation of any $L(x,\dot{x},\ddot{x})$ is fourth order, and the degeneracy escape is closed because the acceleration Hessian of $\mu_{\rm fw}$ is non-singular. **No local higher-derivative action reproduces the law**: the nonlocal form is required, not stylistic. **(4)** Comoving FRW makes u an exact zero mode of $\Box_u$, and $K(0^+) = 0$, so the matter term **vanishes identically on the cosmological background**; since $K \sim \sqrt{z}$ there, no perturbative modified Friedmann equation exists. Modified inertia cannot alter the expansion history. **(5)** The external-field effect is forced to be **quadrature with a vector cross term**, so the customary linear ansatz $A = a_{\rm in} + a_{\rm ex}\theta$ is unavailable; the resulting prediction is a **dipole** rotation-curve asymmetry, attractor-faster, and **independent of the a_0 footing**. **(6)** The linear-response inertia is $h(x) = \mathrm{d}(x\mu_{\rm fw})/\mathrm{d}x = 2x/\sqrt{1+4x^2}$, not $\mu_{\rm fw}$. **(7)** Dispersion-supported systems discriminate the residual closure freedom: the ultralocal member places them **exactly** on the rotation relation, an orbit-averaged member offsets them by -0.037 dex, and on the real McConnachie (2012) catalogue the test proves **systematics-limited** rather than sample-limited (§9.1).

Every result is machine-verified by committed scripts that exit non-zero on any failed internal check.

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1. Scope, and what is not claimed

This paper concerns the **internal structure** of one modified-inertia action. It does not claim to derive the value of a_0, to solve the dark-matter problem, or to establish the framework against alternatives. The author has publicly retracted earlier and broader claims, and nothing here depends on them.

Credit. The interpolating kernel used throughout, $\nu(y) = \sqrt{1+1/y}$, is identical to Milgrom (1999, *Phys. Lett. A* **253**, 273, Eq. 9). The distinctive content of this framework is the **coefficient** $a_0 = cH_\Lambda/Z$ with $Z = \sqrt{32\pi/3}$ (Milgrom's was $2cH_\Lambda$), and the modified-inertia completion. The value of $\kappa = \frac{1}{2}$ in $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ is **not derived**; ghost-freedom, unitarity and holography have each been shown insufficient to force it. This is a one-parameter effective theory.

What five of these theorems do is subtract. They remove a postulate, remove a channel, remove a class of actions, and remove a cosmological sector from the theory's reach. That is the intended contribution: a framework with fewer places to hide is more testable, and the prohibitions were available to any referee.

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2. The action, the closure, and the coefficient

The matter sector carries the modification. With signature $(-,+,+,+)$ and $u\cdot u = -1$,

$$S_{\text{matter}} = -\frac{1}{2}\int\sqrt{-g}\,\rho_m\Big[s\,u^\mu K(\Box_u/a_0^2)u_\mu\Big], \qquad K(z) = \frac{\sqrt{1+4z}-1}{2\sqrt{z}}, \qquad s = -1 \text{ (postulated).}$$

K is Herglotz–Nevanlinna with a unique positive measure, $\|K\| \le 1$, causal-retarded, with $K(0^+) = 0$, $K(\infty) = 1$, and the identity $K(x^2) = \mu_{\text{fw}}(x)$ where

$$\mu_{\text{fw}}(x) = \frac{\sqrt{1+4x^2}-1}{2x}, \qquad x = \frac{|a|}{a_0}.$$

The coefficient. Since $H_\Lambda = \sqrt{8\pi G\rho_\Lambda/3}$ and $Z = \sqrt{32\pi/3}$,

$$\frac{cH_\Lambda}{Z} = c\sqrt{\frac{8\pi G\rho_\Lambda}{3}}\cdot\sqrt{\frac{3}{32\pi}} = c\sqrt{\frac{G\rho_\Lambda}{4}} = \tfrac{1}{2}\,c\sqrt{G\rho_\Lambda},$$

so every π, the 32 and the 3 cancel. Numerically $a_0 = 9.36\times10^{-11}\,\mathrm{m\,s^{-2}}$ on the pure-Λ footing, 1.13×10^{-10} on the ρ_{total} footing. Both are carried through every dimensional statement below.

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3. Theorem 1 — the first moment is worldline-general

Theorem 1. For every timelike worldline, in curved spacetime, with no assumption on the orbit,

$$u_\mu \Box_u u^\mu = -|a|^2 \qquad \Longrightarrow \qquad \langle \Box_u \rangle_u \equiv \frac{u\cdot\Box_u u}{u\cdot u} = +|a|^2.$$

Proof. $\Box_u u^\mu = \dot{a}^\mu$ on a worldline. Differentiate the normalisation twice:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(u\cdot u) = 2\, u\cdot a = 0 \;\Rightarrow\; u\cdot a = 0,$$

$$\frac{\mathrm{d}^2}{\mathrm{d}\tau^2}(u\cdot u) = 2(a\cdot a + u\cdot\dot{a}) = 0 \;\Rightarrow\; u\cdot\dot{a} = -a\cdot a.$$

Dividing by $u\cdot u = -1$ flips the sign, giving $+|a|^2$. $\Box$

Both differentiation steps are verified symbolically for a general worldline with four free functions of proper time. The positivity of the moment — despite the operator's spectrum lying at $-\omega^2$ — comes from the **Lorentzian normalisation** and nothing else.

What is new. The identity was previously checked on circular and exact-helical worldlines. It is in fact an algebraic consequence of $u\cdot u = -1$, hence exact for eccentric, plunging, and perturbed trajectories alike. Everything below uses this generality.

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4. Theorem 2 — non-negativity, and the mutual-exclusivity of MOND and dissipation

Theorem 2. Under the first-moment closure, $\operatorname{Im} K \equiv 0$ identically — for the mean motion and for every perturbation, at every order in perturbation amplitude.

Proof. a^μ is spacelike ($u\cdot a = 0$ with u timelike), so $a\cdot a > 0$ and by Theorem 1 the closure feeds K the argument

$$z = \frac{|a|^2}{a_0^2} \;\geq\; 0.$$

K is real-analytic on $z > 0$ and acquires an imaginary part only on its branch cut, which requires $z \leq -\frac{1}{4}$. The two regions are disjoint. $\Box$

Verified additionally on explicitly perturbed epicyclic worldlines scanned over amplitude $\epsilon \in [0.01, 0.9]$ and frequency ratio $\kappa/\Omega \in [0.2, 3]$: the minimum of $|a|^2$ over every orbit is non-negative, and the argument never approaches $-\frac{1}{4}$.

Corollary 2.1 (mutual exclusivity). The MOND amplitude and any dissipative channel cannot coexist.

- *First-moment closure*: argument ≥ 0, K real, the radial-acceleration relation is reproduced (0.108 dex on SPARC), and dissipation is **identically zero**.
- *Literal spectral closure*: argument $-(\omega c/a_0)^2$ lies on the cut, K is unimodular and complex, dissipation occurs at exactly $a_0/2c$ — but the relation **fails outright** ($K = 0.99999997 + 2.5 \times 10^{-4} i$ against the required $K(1) = 0.618$ at $a = a_0$).

No closure delivers both.

Corollary 2.2 (the sign is demoted). The dissipative rate carries the postulated sign s. Since that rate is identically zero in the closure that reproduces the relation, s has **no observable consequence**: it is not a parameter awaiting better data but a label on a branch the working theory never evaluates. Any future claim to have measured it must first exhibit a closure that samples the cut *and* reproduces the relation — which Theorem 2 forbids within the first-moment family.

Independent confirmation from data. The literal closure's universal drift $a_0/2c = 4.93 \times 10^{-12}\,\mathrm{yr}^{-1}$ ($\tau = 203$ Gyr canonical, 168 Gyr alternative — only a 1.21× footing spread) is excluded at **8.5**σ by the orbital-period agreement of PSR J0737−3039 (Peters back-reaction validated against the measured $\dot{P}_b$ of both J0737−3039 and B1913+16 to 0.03% and 0.8% before use). The Hulse–Taylor system alone reaches only 1.1σ.

Scope. Theorem 2 is proved for the first-moment closure *family*. Within that family the off-circular time-weighting remains free, but every member's argument is a weighted average of $|a(\tau)|^2$ and therefore non-negative, so the conclusion is family-wide.

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5. Theorem 3 — the nonlocal action is necessary

Theorem 3. $\mu_{\mathrm{fw}}(|a|/a_0)\,a = g_{\mathrm{bar}}$ is not the Euler–Lagrange equation of any non-degenerate local Lagrangian $L(x, \dot{x}, \ddot{x})$.

Proof. The higher-derivative Euler–Lagrange equation

$$\frac{\partial L}{\partial x} - \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} + \frac{\mathrm{d}^2}{\mathrm{d}t^2}\frac{\partial L}{\partial \ddot{x}} = 0$$

contains $x^{(4)}$ in general — verified symbolically, highest derivative order 4. The framework's law contains no derivative above $\ddot{x}$; it is an implicit **second**-order relation. The only escape is degeneracy, $\partial^2 L/\partial \ddot{x}^2 = 0$, i.e. L at most linear in $\ddot{x}$. But the framework's dependence enters through $|a|^2 = a\cdot a$ inside the nonlinear μ_{fw}, and the Hessian with respect to acceleration components is non-singular:

$$\det\left[\partial^2 \mu_{\mathrm{fw}}/\partial a_i \partial a_j\right]_{|a|=a_0} = -\frac{11}{25} + \frac{23\sqrt{5}}{125} \neq 0.$$

□

This is a positive result for the published action. The nonlocal form is not an aesthetic preference or a device for evading Ostrogradsky — it is *required*, because no local higher-derivative theory can produce the framework's own phenomenological law.

The gap it exposes. Theorem 1 supplies a **moment identity**: given a worldline, the operator's u-contracted first moment is $|a|^2$. An Euler–Lagrange equation *selects* the worldline. These are different objects, and the corpus's structural guarantees (zero frame degrees of freedom via a no-wave-cone symbol argument, the causal-retarded Herglotz definition, the one-loop sum rule) are theorems about the **nonlocal** action, while every quantitative success is computed with the **local** closure. That the two coincide is not established. The gap is invisible where the framework has been tested: on a circle $|a|$ is constant, so closure members agree to 1.8×10^{-16}, and rotation curves, the a_0-line and the baryonic Tully–Fisher relation are all circular by construction. Off circles the ambiguity grows — 0.07% at $\epsilon = 0.05$, 1.0% at 0.2, 6.5% at 0.5 — and the two places the framework has been pushed off circles are precisely where closure ambiguities appeared.

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6. Theorem 4 — the cosmological background is inert, and non-analytic

> **Theorem 4.** On a comoving FRW background the modified-inertia term vanishes identically, and no perturbative expansion of it exists.

Proof. For $\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\mathrm{d}\mathbf{x}^2$ and comoving $u^\mu = (1,0,0,0)$, all $\Gamma^\mu{}_{00}$ vanish, so

$$a^\mu = u^\nu(\partial_\nu u^\mu + \Gamma^\mu{}_{\nu\lambda}u^\lambda) = 0$$

identically, for arbitrary $a(t)$ — comoving matter is geodesic. Hence $|a|^2 = 0$ and $\Box_u u^\mu = u^a\nabla_a(a^\mu) = 0$: u is an **exact zero mode** of the operator, so by spectral calculus $K(\Box_u/a_0^2)u_\mu = K(0)u_\mu$. Since $K(0^+) = 0$,

$$-\tfrac{1}{2}\rho_m\, s\, u^\mu K(\Box_u/a_0^2)u_\mu = -\tfrac{1}{2}\rho_m\, s\, K(0)(u\cdot u) = 0.$$

Moreover the series at the origin is $K(z) = \sqrt{z}\,(1-z) + \mathcal{O}(z^{5/2})$ with $K'(0^+) = \infty$: K has a **square-root branch point exactly where the cosmological background sits**, so no Taylor expansion in perturbations about it exists. □

Consequences. Modified inertia contributes no background modification and no luminosity-distance tilt — structurally zero, not small. The framework's expansion history therefore comes from its field content, not from the inertia modification. And the coincidence $cH_0/a_0 = 7.00$ is a coincidence of **scales**, not evidence of a **coupling**: nothing in the action makes the expansion rate an argument of K.

The non-analyticity is an obstacle for the perturbation calculation the framework still requires: a naive "expand K to first order in perturbations" step is unavailable because $K'(0^+)$ diverges.

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7. Theorem 5 — the external-field effect is quadrature, and predicts a dipole

> **Theorem 5.** Theorem 1 forces the external field into the closure through
>
> $$|a_\text{in} + a_\text{ex}|^2 = a_\text{in}^2 + a_\text{ex}^2 + 2\,a_\text{in}\cdot a_\text{ex},$$
>
> a **quadrature** sum with a **vector** cross term. No scalar phase function multiplying a_ex can appear.

The customary linear ansatz $A = a_\text{in} + a_\text{ex}\theta(\omega_\text{ex}/\omega_\text{in})$ is therefore unavailable in this framework, together with everything derived from it.

The correct construction. The equation of motion applies to the total force and total acceleration,

$$\mu_\text{fw}\!\left(\frac{|a_\text{tot}|}{a_0}\right) a_\text{tot} = g_\text{bar,in} + g_\text{ex},$$

the host's centre of mass obeys the same law under g_ex alone, and the observable is the difference $a_\text{in,obs} = a_\text{tot} - a_\text{ex}$, because the host frame accelerates. (Omitting the host's own acceleration double-counts the external field and inflates the effect several-fold; the isolated limit of the construction above reproduces the framework's relation to machine precision.)

Corollary 5.1 (the dipole). Around a circular orbit a_{in} rotates while a_{ex} is fixed, so the cross term modulates the inertia with azimuth. On the side facing the attractor the fields partially cancel, $|a_{\text{tot}}|$ falls, μ_{fw} falls, and the boost rises: the **near side rotates faster**. The prediction is a **dipole**, not a quadrupole, and it is **independent of the a_0 footing**, because the governing equation is written entirely in units of a_0 and the amplitude depends only on $e = g_{\text{ex}}/a_0$ and $y = g_{\text{bar}}/a_0$.

regime	dipole $(v_{\text{near}} - v_{\text{far}})/\bar{v}$
$e \ll y$ (inner disk, weak external field)	4.2% – 22.3%
$e \sim y$ (**MOND saddle — treatment invalid**)	4.2% – 41.1%
$e \gg y$ (strong external field)	0.1% – 1.2%

The $e \sim y$ peak is the known MOND saddle, where $g_{\text{tot}} \to 0$ on the near side and the interpolation sits at its singular point. That is real physics but a point-particle balance is not valid there; a proper field solve is required. The $e \gg y$ limit correctly **erases** the dipole, everything being Newtonianised — a check on the construction.

A prior claim is retracted. It has been stated in this corpus that pure modified inertia predicts *exactly zero* directional asymmetry. That is **wrong**, and its origin is now clear: a scalar θ multiplying a_{ex} carries no orientation, so the asymmetry vanishes identically under the borrowed ansatz. Quadrature has a vector cross term, so orientation survives. Away from the saddle the framework predicts 4.2%–33.6%, *larger* than the 1–4% of aligned-rotation-curve asymmetry expected in AQUAL-type modified gravity. The observable therefore changes character: from blind to modified inertia, to **favourable** to it, as an amplitude comparison.

Falsifier. A measured rotation-curve dipole of the **opposite** sign (far side faster) above $\sim 1\%$. The sign is forced by the cross term and cannot be tuned.

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8. Proposition 6 — the linear-response inertia

For $F = m\,a\,\mu_{\text{fw}}(a/a_0)$, the response to a small additional force is governed by the derivative, not the ratio:

$$h(x) \equiv \frac{\mathrm{d}}{\mathrm{d}x}\left[x\,\mu_{\text{fw}}(x)\right] = \frac{2x}{\sqrt{1+4x^2}},$$

so the amplification of a perturbing response is $1/h(x)$, **not** $1/\mu_{\text{fw}}(x)$. At $x = 1$ these are 1.118 and 1.618; in the deep regime they differ by a factor 2 ($1/h \to 1/2x$ against $1/\mu_{\text{fw}} \to 1/x$). Conflating them overstates perturbative responses.

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9. Proposition 7 — dispersion-supported systems discriminate the residual closure freedom

The off-circular time-weighting of $|a(\tau)|^2$ is free within the first-moment family. Two members:

- **Ultralocal:** $x_A(\tau) = |a(\tau)|/a_0$, applied pointwise.

- **Orbit-averaged:** $x_B = \sqrt{\langle |a|^2 \rangle_{\text{orbit}}}/a_0$, one number per orbit.

On a circle $|a|$ is constant and they coincide, which is why rotation curves cannot separate them. Dispersion-supported systems can.

> **Proposition 7.1.** Under the ultralocal member, spherical dispersion-supported systems lie **exactly** on the rotation relation, for any orbit shape, because force balance is pointwise algebraic. Verified to 1.6×10^{-15} across four decades in g_{bar}, both footings.

> **Proposition 7.2.** Under the orbit-averaged member they are offset **below** it. The sign follows before computation: x_B is constant along an orbit while x_A varies and is smallest at apocentre, where stars spend most of their time; so $\mu_{\text{fw}}(x_A) < \mu_{\text{fw}}(x_B)$, and since $\mu_{\text{fw}}(x)a = g_{\text{bar}}$ the orbit-averaged closure over-assigns μ and under-predicts the acceleration.

Orbits integrated in the deep-regime logarithmic potential give, by apocentre-to-pericentre ratio k: -0.015 dex ($k = 2.2$), -0.034 ($k = 3.7$), -0.062 ($k = 8.3$), -0.088 ($k = 20.5$); representative mean over $k \sim 2$–4 is $-$**0.037 dex**.

This does not yet fix the weighting — -0.037 dex hides inside a 0.15 dex detectability tolerance and below the framework's own 0.11 dex relation scatter, so the whole family survives.

9.1 Confrontation with the real catalogue (new in v2)

The test was run on **McConnachie (2012)**, AJ **144**, 4 (VizieR J/AJ/144/4/catalog): 46 of 102 rows carry a stellar velocity dispersion, a half-light radius and an absolute V magnitude. Using the Wolf et al. (2010) estimator with $r_{1/2} = \frac{4}{3}R_e$, $g_{\text{obs}} = 3\sigma^2/r_{1/2}$ and $g_{\text{bar}} = G(\Upsilon_V L_V/2)/r_{1/2}^2$, residuals taken against this framework's own ν:

quantity	v1 assumption	**real catalogue**
per-object scatter	0.15–0.20 dex	**0.38–0.48 dex**
dominant error	random per object	**coherent** (stellar Υ_V)
mean residual, $N = 29$ after pre-stated cuts	—	$+0.2945 \pm 0.0710$ dex (random only)

The forecast fails on both counts. The scatter is 2–3× larger than assumed, and — decisively — the dominant uncertainty is not random. Υ_V is common to the whole sample, so it displaces every dwarf identically and $\sqrt{N}$ leaves it untouched. Varying Υ_V over 1–4 moves the mean residual from $+0.454$ to $+0.132$: a span of **0.322 dex, 8.7× the 0.037 dex signal**. Matching the signal would require Υ_V known to 0.074 dex (19%), against a literature spread of 50–100%.

Consequently **“**$N \sim 150$ **for 3 ” and “both routes are archival” are withdrawn.** The measurement is systematics-limited, and more dwarfs do not help.

9.2 The route that survives

A coherent Υ error and a closure offset are degenerate only if they share a g_{bar} dependence. On the real sample they do not: $\partial(\text{residual})/\partial \log \Upsilon$ has a slope of -0.067 per dex against $\log_{10}(g_{\text{bar}}/a_0)$, while the closure offset is approximately flat in the deep regime — over an available **5.28 dex** of dynamic range. So the two are **partially separable in principle.**

Realising that requires the closure offset computed **across** g_{bar}, whereas §9 above evaluates a single deep-regime orbit family. That is the concrete next step and it is not done here. The alternative is a sample with independently calibrated stellar masses.

Scope. Planar orbits in an idealised logarithmic potential; three representative orbits rather than a self-consistent distribution function. Sign and order are robust; the precise dex value should be read as order-of-magnitude, and a proper isotropic and anisotropic distribution function is the next step.

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10. What this changes

item	before	after
first-moment identity	verified on circles/helices	**worldline-general** (Thm 1)
dissipative channel	open, sign postulated	**identically zero**; sign demoted (Thm 2)
choice of nonlocal action	a modelling preference	**required** (Thm 3)
cosmological background	"unbuilt"	**structurally inert**, and non-analytic (Thm 4)
external-field effect	Milgrom's postulated θ	**derived quadrature**; footing-free dipole (Thm 5)
perturbative response	$1/\mu_{\mathrm{fw}}$	$1/h$, differing by up to $2\times$ (Prop 6)
off-circular weighting	free $O(1)$ choice	**predictions sharpened** (Props 7.1, 7.2); the test is **systematics-limited**, needing Υ_V to 19% or the g_{bar}-resolved offset (§9.1–9.2, **v2**)

Two remain untouched and postulated: the **value** of $\kappa = \frac{1}{2}$, and the coefficient Z.

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11. Reproducibility

Every number is produced by a committed script that exits non-zero if any internal check fails. No verdict is hard-coded.

script	results
`mi_dcac_split_settled_2026.py`	Thms 1, 2; Cors 2.1, 2.2
`mi_sign_from_perturbation_drift_2026.py`	the 8.5σ pulsar exclusion; the universal rate
`mi_closure_vs_action_gap_2026.py`	Thm 3 and the moment-vs-EL gap
`mi_channelA_friedmann_2026.py`	Thm 4
`mi_efe_derived_general_2026.py`	Thm 5, Cor 5.1
`mi_growth_amplification_founded_2026.py`	Prop 6
`mi_closure_fixed_by_rar_universality_2026.py`	Props 7.1, 7.2
`mi_dsph_closure_test_real_data_2026.py`	§9.1–9.2, the real-catalogue confrontation (**v2**)

script	results
`mi_kappa_spectral_reduction_2026.py`	the κ reduction of §2

References

- M. Milgrom, *Phys. Lett. A* **253**, 273 (1999) — the $\nu(y) = \sqrt{1 + 1/y}$ kernel.
- M. Milgrom, arXiv:2208.07073 — the external-field effect and the θ ansatz.
- M. Kramer *et al.*, *Phys. Rev. X* **11**, 041050 (2021) — PSR J0737−3039 timing.
- J. Weisberg & Y. Huang, *Astrophys. J.* **829**, 55 (2016) — PSR B1913+16 timing.
- P. C. Peters, *Phys. Rev.* **136**, B1224 (1964) — orbital back-reaction.
- Framework antecedents: Zenodo 10.5281/zenodo.21264727 (action v4), 10.5281/zenodo.21284144 (one-loop edge), 10.5281/zenodo.21702746 (wide-binary gate law).